

ADNet: A Neural Network for Accelerometer Signals Denoising

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Abstract—Accelerometer signals play a critical role in many fields, for example navigation and vehicle safety. However, uncontrollable factors such as defective equipment and harsh environments make the signals recorded by sensors contain a large amount of noise, which poses a great challenge. Most existing methods are based on traditional signal processing and often face problems of incomplete noise reduction or signal distortion after denoising. In this paper, we propose a data-driven denoising method for accelerometer signals (ADNet) based on the Wave_U_Net network architecture, incorporating Multi-Head Attention mechanism and Spatial Attention mechanism. This enhances the network's feature extraction capability and accelerates its convergence speed. Meanwhile, the attention mechanism focuses the model's attention on the clean signal's feature information, improving the network's fitting capability to the signal distribution. Therefore, ADNet can achieve more thorough denoising while reducing signal distortion after denoising. Finally, we conduct experiments on Walking speed dataset and field experiment dataset to verify the effectiveness of ADNet. The experimental results show that ADNet outperforms other baseline models in terms of Mean Square Error, Mean Absolute Error, Root Mean Square Error, and Signal-to-Noise Ratio.

Index Terms—Accelerometer Signals, Denoising, Convolutional neural network, Attention mechanism

I. INTRODUCTION

Accelerometer sensors, as a type of universal sensor, can convert the displacement and speed information of an object into a digital signal in real time and transmit it to the cloud platform or other devices through the network, providing core data support and a decision basis for monitoring the status of the object through the devices. In practice, however, sensors are susceptible to harsh environments and defects in their equipment, resulting in recorded accelerometer signals containing large amounts of noise. For example, Surrounding electrical equipment and wiring may cause electromagnetic interference to the sensor [1]; molecular movements and charge rise and fall within electronic components can generate thermal or random noise [2]. The integration of noise measurement presents significant challenges for subsequent high-precision work, making it crucial and valuable to research denoising accelerometer signals.

Traditional algorithms, such as Fast Fourier Transform, Wavelet Transform, Kalman filtering, and Empirical Mode

Decomposition, are commonly used to denoise the output signals of sensors [3]. These traditional algorithms require manual selection and adjustment of filter parameters, which is tedious and requires specialized knowledge. Improper selection can result in excessive or incomplete denoising. With the development of artificial intelligence, neural networks have played a significant role in various fields [4]–[7]. Therefore people began to study the combination of neural networks and traditional algorithms to improve their shortcomings. For example, [8] utilizes neural networks to assist in selecting appropriate thresholds for wavelet transforms. While such algorithms can enhance the performance of traditional methods, their underlying algorithm remains traditional algorithms.

New data-driven denoising algorithms can automatically learn the features of signals and the distribution of noise based on large amounts of training data, and can adapt to changing noise environments. They have achieved significant results in fields such as speech denoising [9]. However, to the best of our knowledge, their application in accelerometer signal denoising is currently rare. The accuracy of acceleration is crucial in various fields, such as motion tracking and object localization [10]. Therefore, in this paper, we propose a new neural network model that can successfully eliminate noise in the accelerometer signal output by sensors, while also reducing distortion of the denoised signal. The main contributions of this paper are summarized as follows:

- We propose a new neural network model (ADNet) to address the issue of poor denoising effectiveness in existing methods for accelerometer signal denoising.
- We introduce Multi-Head Attention and Spatial Attention based on the Wave_U_Net. This focuses the attention of model on the feature of clean signals, alleviating distortion in the signal after denoising, and achieving better denoising.
- We conduct experiments to validate the denoising effect of ADNet on both the Walking speed dataset and field experimental dataset. The results show that ADNet outperforms other baseline models.

The rest of the paper is as follows: Section II describes the

related work on the denoising methods for the output signals of sensors. Section III presents our proposed solution approach, Section IV elaborates on the data preparation process, Section V shows the experiments and analysis of the results, and Section VI presents the conclusion.

II. RELATED WORK

We divide this section into two parts according to the adopted solution method: the traditional methods and the data-driven based methods.

A. The Traditional Methods

In digital signal processing, traditional denoising methods mainly include simple filtering algorithms, Kalman filtering, and Wavelet-based algorithms. [11]–[13] used different filters to simply filter the original signal to achieve the purpose of denoising. This type of algorithm has a simple structure, is easy to implement, and has low computational cost during algorithm execution. However, finding the optimal gain vector of the filter in this algorithm is a challenge and there is a problem of statistical bias. The Kalman-based algorithm is a linear system algorithm proposed by Kalman for estimating random state processes and can be used to estimate the noise in system measurements [14]. Since real noise is complex, and diverse, many researchers [15] have proposed improvement strategies. Nevertheless, the essence of the Kalman algorithm is to estimate measurement noise, and the complexity and diversity of real noise pose great difficulties for algorithm estimation. The algorithm itself needs to establish an error model and is usually used in conjunction with other system algorithms. Once a subsystem fails, it will seriously affect the accuracy of the algorithm. Moreover, the uncertainty of model parameters can cause the algorithm to fail to converge normally, ultimately leading to poor denoising performance of the algorithm. The algorithm based on wavelet transform does not need to establish an error model [16], and it can perform multi-resolution analysis of signals in both time and frequency domains. However, wavelet transform has poor adaptability to complex real noise, and it usually needs to cooperate with other algorithms to achieve a certain denoising effect [17]. Although predecessors have done a lot of research on the determination of threshold value and the selection of wavelet basis in this algorithm [18], it is difficult to find the ideal threshold value. Most of them are determined by empirical knowledge and lack reasonable theoretical basis, which increases the difficulty of using this algorithm.

B. The Data-driven Based Methods

With the development of neural network algorithm, it has been widely used in sequence problems, such as speech recognition [19]. The vanilla Recurrent Neural Network (RNN) is a fully connected neural network with recurrent feedback. The information at each time in the network contains the information at the previous time, so the network can well capture the time series information in sequence data. The output signal of accelerometer sensor can often be regarded

as a time series signal, so a study [20] apply RNN to sensor data denoising. However, due to the characteristics of RNN structure, the gradient disappearance phenomenon is prone to occur, and this phenomenon often occurs on the time axis, so RNN is limited by long-term dependence. The Long Short-Term Memory (LSTM) network is used in the study [21], which selectively stores the information of the hidden layer by adding a gating unit, so as to achieve the control of long-term memory. Gated Recurrent Unit (GRU), another variant of RNN, is used to solve the shortcomings of RNN like LSTM, but the difference is that LSTM chooses to expose part of the hidden layer information, while GRU chooses to expose all the information. In comparison, GRU has fewer network parameters and faster convergence speed. Research [22] applied the variants of these three network models to the denoising of accelerometer signals, and achieved better denoising results compared with the traditional denoising methods.

Different from the above RNN, convolutional neural network (CNN) is used to complete the corresponding task in the data of accelerometer sensors [23]. Convolutional kernels in the CNN help the network achieve parameter sharing, greatly reducing the number of parameters in the network. U-Net as a variant of CNN was initially applied in the field of medical image segmentation [24], and is now commonly used in various fields. Some studies [25] found that it could also play a significant role in image denoising tasks, so some studies [26] introduced it into speech enhancement tasks. Since speech can be represented as one-dimensional waveform data, this study converts the two-dimensional convolution in the network into one-dimensional convolution, and replaces the transposed convolution in the original network by using linear interpolation and normal convolution to ensure the continuity of the feature information learned by the network.

From most studies, it can be easily found that there are few studies that apply neural networks to data of accelerometer sensors [22]. Accelerometer devices typically provide data for the X-axis, Y-axis, and Z-axis. However, the data on each axis is a one-dimensional time series similar to the waveform structure of speech data. Therefore, we propose a new network structure, which uses Wave-U-Net network architecture as the basic network model and combines the attention mechanism to make up for the lack of convolutional neural network in obtaining global feature information to complete the denoising task of the accelerometer signals.

III. ADNET

This section describes a neural network model that we propose for denoising accelerometer signals. A noisy signal can be generally represented as $Y = X + N$, and our model is trained to map the input sequence of noisy signals $(y_1, y_2, \dots, y_n)$ to a sequence of clean signals $(x_1, x_2, \dots, x_n)$.

A. The Base Architecture

A schematic of the structure of our model is shown in Fig.1. It consists of an encoder and a decoder. The encoder

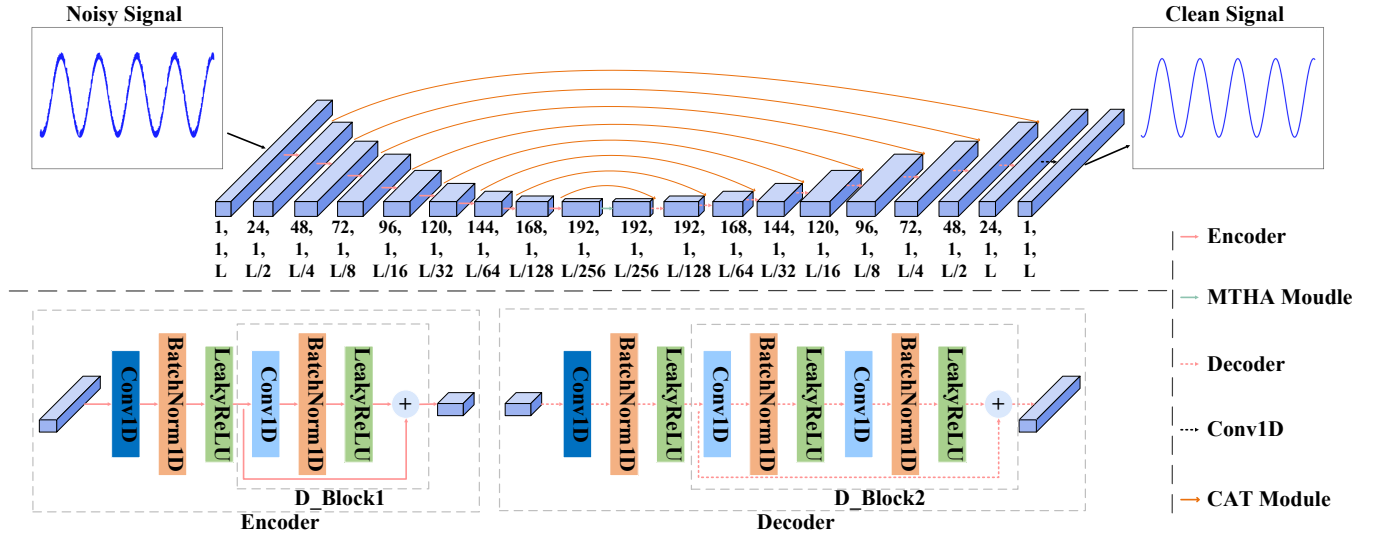


Fig. 1. Overview of ADNet structure. ADNet uses an 8-layers encoder and decoder. Skip connections are applied between the corresponding layers of the encoder and decoder using CAT module to provide crucial spatial feature information. The intermediate layers between the encoder and decoder are connected using MTHA module to learn global feature information.

Algorithm 1: The ADNet Algorithm

Input: Noisy accelerometer signal X_n ,
Clean accelerometer signal X_c ;
Output: Denoising accelerometer signal Y_d ;

- 1 Normalize the data to $[0, 1]$;
- 2 */* train */*
- 3 **for** number of training iterations **do**
- 4 **for** i of training batchsize **do**
- 5 $Y_{d_i} = f_{\theta}(X_{n_i})$;
- 6 $L(Y_{d_i}, X_{c_i}) = \frac{1}{batchsize} \sum (Y_{d_i} - X_{c_i})^2$;
- 7 Update $\theta \leftarrow \theta - \alpha \nabla_{(\theta)} L$;
- 8 */* $f_{\theta}(X)$: our model function;*
- 9 *θ : the parameters of our model;*
- 10 *$L(Y_d, X_c)$: our Loss function-MSE Loss; */*
- 11 */* test */*
- 12 **for** number of training iterations **do**
- 13 Get $Y_d \leftarrow$ Input the new accelerometer signal into the Model repeat Steps 4,5;
- 14 Compare Y_d and X_c to computer evaluation metrics;
- 15 **return** : The final denoising accelerometer signal Y_d ;

compresses the data layer by layer, calculating the feature information of the data at different time scales, which helps the model to thoroughly remove noise. The decoder restores the feature information through upsampling, and finally obtains the predicted clean signal. The upper and lower layers of the encoder and decoder are directly connected through the D_Block module, which enables the network to fully utilize the feature map of each layer, thereby slowing down the gradient vanishing and accelerating network convergence. The

encoder jumps to the corresponding layer of the decoder through the channel attention (CAT) module, providing key channel feature information for the decoder. The intermediate layer between the encoder and decoder is an Multi-Head Attention (MTHA) module, which enhances the model's ability to learn global feature information and helps to reduce the distortion of the denoised signal.

The main framework of our encoder and decoder in the architecture is similar to that of the Wave-U-Net network. We have adjusted it to have 8 layers. This is because we focus the model's attention on clean data by introducing the CAT and MTHA modules, which have the attention mechanism as their main component, to improve the model's ability to fit clean data distributions and achieve reduced signal distortion after denoising. This strategy introduces a large number of parameters into the model, which can make it significantly more difficult to train the model. Therefore, the depth of the model is moderately adjusted to achieve a trade-off.

However, drastically reducing the depth of the model would lead to a reduction in the model's ability to extract feature information from the data, so we introduce two versions of the D_block module, which consists mainly of Conv1D, BatchNorm1D, LRelu, and residual connections. We introduce the D_block1 module in each layer of the encoder can help the network to fully extract and utilize the feature map of each layer and speed up the network convergence. At the same time, introducing the D_block2 module in each layer of the decoder, which can help expand the receptive field of the network and improve the fitting ability of the network, thus helping the decoder to restore the data better. The structure of each layer of the encoder and decoder is shown in the Fig.1.

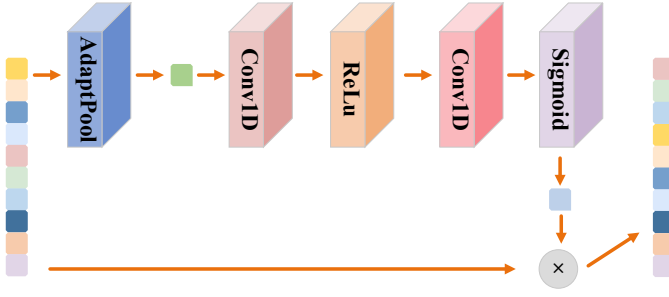


Fig. 2. Overview of CAT Module.

B. Attention Module

As can be seen from the previous subsection, decoder restores data through upsampling, and the upsampling operation requires lots of information to fill the gaps, so the network adopts skip connection to supplement the spatial feature information required by the model. However, skip connection can not make up for the discontinuity of feature information caused by upsampling, thus linear interpolation and normal convolution are used to replace upsampling in Wave-U-Net. Although this ensures the continuity of feature information, linear interpolation will introduce noise that does not exist in the original data to interfere with model training. Therefore, inspired by [27], we add CAT module to the skip connection part of each layer of the network. We replace all feed forward operations with convolution operations to reduce the number of parameters of the network, and its structure is shown in Fig.2.

The entire network is trained under the guidance of the loss function, and the learned channel coefficients make the network pay more attention to channels containing important features while ignoring channels containing more noise. This can suppress the impact of noise introduced by linear interpolation on the denoising performance of the network, and also preserve the continuity of feature information provided by linear interpolation, thus providing more clear and important spatial feature information to the decoder.

The effect of convolution operation in the visual field is significant, because the image pixel information has translation invariance, which makes the convolution operation extract local features and learn image features more quickly and accurately. However, our acceleration data is a time series data, and the data at the previous moment has a great influence on the characteristics of the following data. So the global and temporal step information is crucial for the overall network to learn the features of the data. Combined with the Transformer [28], we introduce the MTHA module into the original network structure to improve the ability of the model to learn global feature information. The structure diagram of this module is shown in Fig.3.

We consider the fusion of MTHA modules at three locations in the network. (1) Skip connection (2) Encoder or Decoder (3) The Middle layer. We finally choose (3). The reasons for this choice are as follows. First of all, due to the length of our data,

if MTHA module is introduced in the upsampling, downsampling and jump connection stages, network parameters will be greatly increased, which not only increases the training difficulty of the network, but also reduces the performance of the network. Secondly, the purpose of upsampling is mainly to restore the data, and the purpose of jump connection is to provide spatial features for upsampling, which is not consistent with our purpose of introducing the MTHA module. The middle layer is the bridge between downsampling and upsampling. Downsampling extracts data features layer by layer to fully extract local feature information of multiple time scales of data. This feature information is then passed into the middle layer of the fusion MTHA module, which can help the network learn the correlation strength between each feature in the feature information and other features, and transmit the data highlighting the key feature information into the upsampling stage, and then assist the recovery of data features with focus in the upsampling stage, thus enhancing the denoising effect of the model. Algorithm 1 shows our ADNet with pseudo-code in detail.

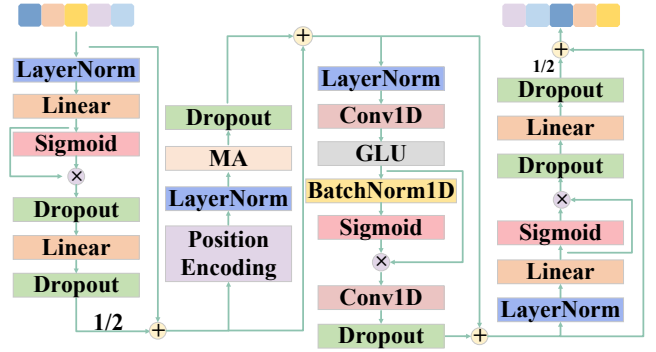


Fig. 3. Overview of MTHA Module. MA stands for multi-headed self-attention with relative positional embedding. GLU is the gated linear unit.

IV. DATASETS

A. Walking speed Dataset

Excluding external environmental factors, mechanical noise contains an inherent intrinsic mechanical noise mechanism attributed to Brownian motion. This is caused by dynamic unbalanced forces resulting from molecules interacting with small particles or experiencing random shocks, also known as random walk noise [29]. MEMS accelerometers are primarily subjected to short-term random walk noise and long-term zero-bias instability error noise. Random walk noise presents itself as high frequency signals and can be effectively characterized by white noise, while zero-bias instability noise can be described as exhibiting $1/f$ noise or other low-frequency drift. Therefore, we simulate the output signal of MEMS accelerometer as Eq. (1).

$$f(t) = x(t) + noise_1(t) + noise_2(t) \quad (1)$$

Where, $f(t)$ is the output signal of the accelerometer. $x(t)$ is the original accelerometer signal. $noise_1(t)$ is a white Gaussian noise (AWGN) with a mean of zero and a variance of σ^2 . $noise_2(t)$ is the noise of $1/f$. We use the X-axis accelerometer signal in the dynamic model of pedestrian acceleration when walking constructed by [30] as the original accelerometer signal in our simulation data, and the Eq. (2) is as follows.

$$x(t) = -\beta \sin(2\pi ft) \quad (2)$$

Where, β is a constant value and f is the sampling frequency of the signal.

We generate the required simulated dataset based on the model formulation described above. When generating data in each time period, a value of β is uniformly chosen from $[0.5, 1]$, f is set to 50Hz , the Hurst exponent in the noise of $1/f$ is $H = 0.01$, σ^2 in a Gaussian white noise is set to 0.02^2 , and we control the addition of 70% noise to the original accelerometer signal by setting a parameter $K = 0.7$.

B. Field Experiment Dataset

The field experimental dataset is from [22], which is obtained by processing the dataset published in [31]. A special mechanical setup was constructed in the collection device for this dataset. This device allows the HuaWei P40 smartphone to be aligned with the MRU unit of the Inertial Laboratory, setting their sampling rate to the same, 100Hz, and then synchronously starting to work to collect signal data. In this way, we treat the low-performance measurements collected by the Huawei P40 smartphone as noisy signals in our experiments; The ground truth (GT) collected by the MRU unit is used as the original clean signal in our experiment. They form our dataset as a whole.

V. EXPERIMENTS

We evaluate the performance of ADNet on simulated and experimental datasets and compare it with previous methods. To validate the effectiveness of the proposed strategy, we conduct ablation experiments. In the following sections, we will describe the implementation details.

A. Experimental Set

Our proposed architecture requires no pre-training and it can be trained through an end-to-end strategy. In the training process, we set the length of a sample $L = 4096$, and the input of the model is noisy-clean data pairs. Before the data is inputted into the model, we standardize it to reduce the difficulty of model training. We use Adam optimizer with momentum term (0.9, 0.999), the initial learning rate is set to $1e-3$, and MSE Loss is used as the loss function of the network.

In order to demonstrate the effectiveness of the proposed model, we take the traditional method (DWT), the variants of LSTM, RNN and GRU proposed in [22], and the two models TSTNN [32] and CAUNet [33], which have significant effects in the field of speech enhancement, as the baseline. To be fair, epochs are set to 1200 for all models. All experiments

are implemented on PyTorch and trained on a single NVIDIA 3080Ti GPU.

In the experiment, the final denoising effect is evaluated by four indicators: Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean squared Error (RMSE) and Signal-to-Noise Ratio (SNR). The four indicators are defined as in Eq. (3-6).

$$MSE(y_e, y_c) = \frac{1}{m} \sum_{i=1}^m (y_{e_i} - y_{c_i})^2 \quad (3)$$

$$MAE(y_e, y_c) = \frac{1}{m} \sum_{i=1}^m |y_{e_i} - y_{c_i}| \quad (4)$$

$$RMSE(y_e, y_c) = \frac{1}{m} \sum_{i=1}^m \sqrt{(y_{e_i} - y_{c_i})^2} \quad (5)$$

$$SNR(y_e, y_c) = 10 \log_{10} \frac{\sum_{i=1}^m y_{c_i}^2}{\sum_{i=1}^m (y_{e_i} - y_{c_i})^2} \quad (6)$$

Here, y_c represents the real clean signal, and y_e represents the denoised signal output by the model. Among these indicators, lower MSE, MAE and RMSE indicate better denoising effect, while higher SNR indicate less noisy signals.

B. Results on Walking Speed Dataset

TABLE I
DENOISING COMPARISON: Walking

	RMSE[m/s ²] $\times 10^{-3}$	RAE[%]	MAE[m/s ²] $\times 10^{-3}$	SNR[dB]
Noisy	28.000	4.824	24.000	26.60
DWT	7.000	1.411	6.000	37.18
Wave-U-Net	0.577	0.94	0.453	60.53
TSTNN	3.500	0.642	0.800	44.58
CAUNet	1.612	0.299	0.700	50.54
LSTM-v	19.120	3.123	16.010	30.58
RNN-v	4.572	0.747	3.794	43.18
GRU-v	4.774	0.778	3.950	42.83
Our-ADNet	0.299	0.049	0.221	66.24

In the simulated dataset, we compare the denoising effect of the proposed model with that of the baseline model, and the results are presented in Table I. It is obvious that our proposed method can obtain the highest SNR (66.24) value and the lowest value of RMSE (2.998×10^{-4}), RAE (0.049), MAE (2.216×10^{-4}).

DWT relies on error analysis model, which greatly limits the ability of the algorithm, while data-driven methods can directly learn the mapping between data, so the denoising effect is generally better than traditional methods. Although the algorithm based on RNN can fully consider the time dimension of data, it is also affected by long-distance dependence. TSTNN and CAUNet models are mainly applied in the field of speech enhancement. Although the waveform structure of speech is similar to our acceleration data, it is not the same as most noise in speech. TSTNN removes the position encoding because of the acoustic feature, which is an indispensable feature in the accelerometer signal, so their denoising effect on the

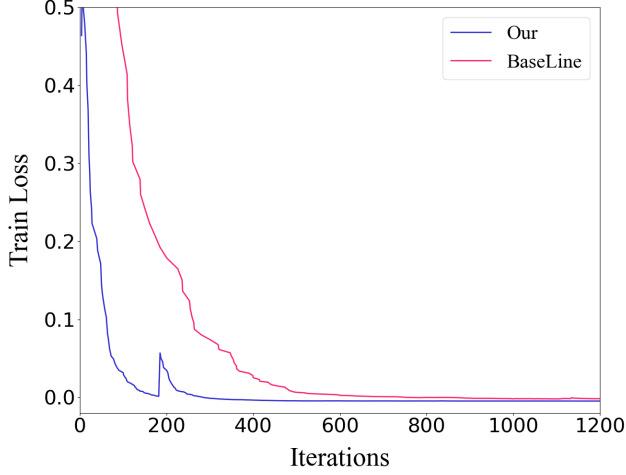


Fig. 4. Loss curve of network training.

accelerometer signal is not satisfactory. Our model obtains high-dimensional features at multiple time scales through multiple layers of convolutional structures, and uses attention mechanisms to capture global dependencies and concentrate the attention of network on key feature information, which can not only achieve the denoising effect but also reduce the degree of distortion after noise reduction.

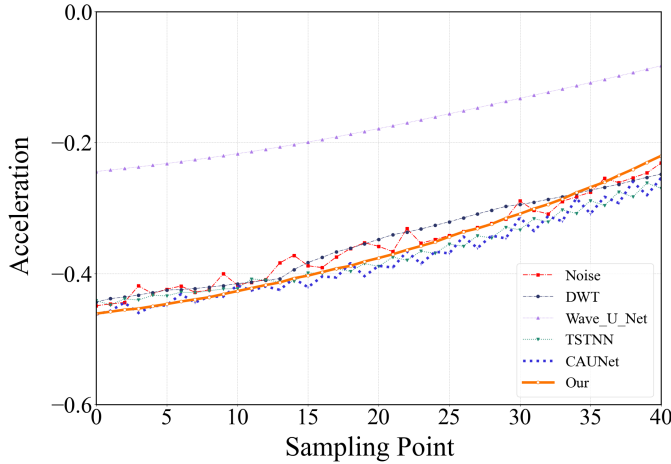


Fig. 5. Denoising comparison: simulation.

We plot the denoising results of a noisy signal of the traditional method DWT and the data-driven method whose denoising effect is ranked in the top four according to the index SNR in the Fig.5. From the figure, we can easily observe that the result curves obtained by DWT, TSTNN and CAUNet methods are not smooth, indicating that the denoising effect is not complete. On the contrary, the result curves obtained by Wave-U-Net and our proposed method are relatively smooth, which indicates better denoising effect. In this simulated dataset, the curve of the clean signal and the smooth Noise curve are nearly overlapping, in close

value intervals. However, there is a large gap between the curve obtained by Wave-U-Net and the Noise curve, which means that the final denoising result obtained by this method is accompanied by large signal distortion. The result curve obtained by our proposed method is not only smooth but also close to the Noise, which indicates that our model can not only denoise well but also ensure that the signal distortion after denoising is less. In addition, the modifications to our model based on Wave_u_net also accelerate the convergence of the model, as shown in the Fig.4.

TABLE II
DENOISING COMPARISON: Field Experiment

	RMSE[m/s ²]	RAE[%]	MAE[m/s ²]	SNR[dB]
DWT	0.1030	4.02	0.0980	29.09
Wave-U-Net	0.0041	0.71	0.0032	43.30
TSTNN	0.1220	7.80	0.1150	30.80
CAUNet	0.1190	7.59	0.1130	30.84
LSTM-v	0.3160	54.16	0.2760	5.386
RNN-v	0.1750	30.34	0.1400	10.62
GRU-v	0.2770	47.32	0.2340	6.58
Our-ADNet	0.0038	0.64	0.0029	44.09

C. Results on Experimental Dataset

In the experimental dataset, we make the same experimental comparison as in the simulation dataset (Table II). The performance of each model is reduced for two main reasons. On the one hand, noise in reality is complex and diverse unlike noise added in simulated data sets which has a fixed data distribution. On the other hand, with the dataset length $L=200$ in the public dataset [22], we made the same splicing process between the noise data and GT data to obtain the data with the length $L=4096$, but this would cause data discontinuity in a sample and bring greater challenges to the denoising of the model. However, under the same data conditions, our model can still achieve the best indicators.

D. Ablation Study

In order to demonstrate that the proposed model can effectively improve the denoising effect, we use the 8-layers Wave-U-Net model as the benchmark architecture to conduct six sets of tests on different combinations of the three proposed improved structures. The denoising effects of different improvement strategies are shown in Table III.

TABLE III
DENOISING COMPARISON: Ablation

	RMSE[m/s ²] $\times 10^{-4}$	RAE[%]	MAE[m/s ²] $\times 10^{-4}$	SNR[dB]
Base model	9.210	0.17	7.878	56.08
Base+D_Block	7.377	0.14	5.483	58.13
Base+CAT	8.126	0.15	6.177	57.26
Base+MTHA	6.171	0.12	5.018	59.69
Base+D_Block+CAT	6.386	0.12	4.956	59.18
Base+CAT+MTHA	6.558	0.12	5.023	59.21
Base+D_Block+MTHA	5.111	0.09	3.936	61.31
Our-ADNet	2.998	0.05	2.216	66.24

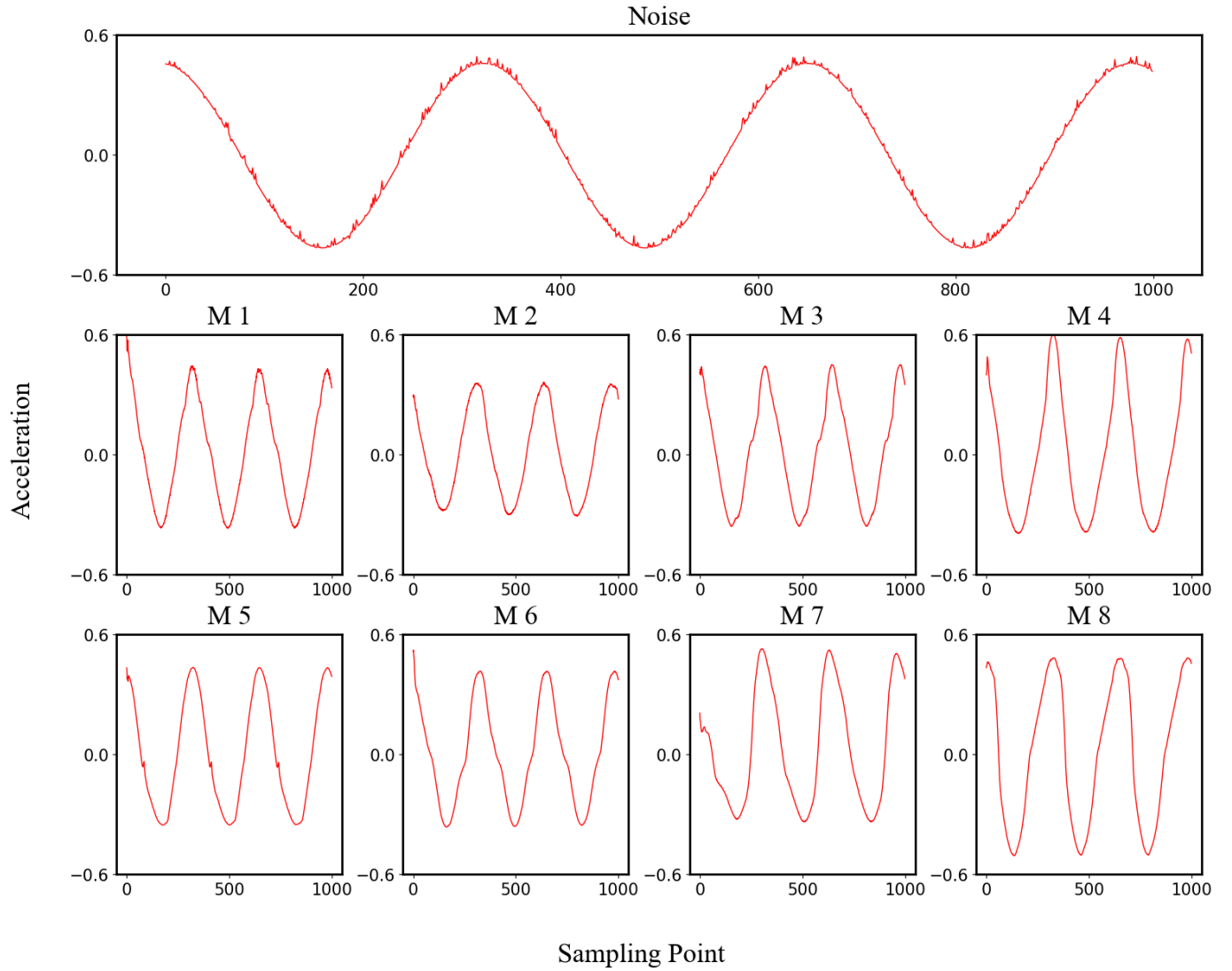


Fig. 6. Denoising effect of different improvement strategies. M1 to M8 are arranged in the following order: base model, base_Dblock model, base_CAT model, base_MTHA model, base_D_block_CAT model, base_CAT_MTHA model, base_Dblock_MTHA model and our model.

Through ablation studies, we can see that the denoising effect of all the improved strategies is better than that of the basic framework model, and our proposed method is again better than all six improved strategies, obtaining the lowest RMSE, RAE, MAE and the highest SNR. It can be seen from various indicators that although the denoising performance of the model using the channel attention module is less than that of other improved programs, the final denoising effect after fusion is improved, indicating that this module plays a supporting role. To further clearly see the impact of each improvement strategy on the results, we plot their partial denoising results for a noisy signal in Fig.6.

It can be seen from Fig.6 that although M1 has denoising effect, it has very limited denoising effect in the turning section of the signal. Although the denoising effect of M2 is better than that of M1, it still suffers from the same problem. The

denoising effect of M3-M7 is better than the first two models (M1-M2), but there is more or less signal distortion. M8 retains the characteristics of the original signal to a greater extent, and reduces the distortion degree of the denoised signal while denoising, which confirms the effectiveness of each module of our model.

VI. CONCLUSION

In this paper, we apply artificial intelligence to the field of mechanical engineering. We propose a neural network model, ADNet, for denoising accelerometer signals. ADNet primarily uses an encoder to learn multi-time-scale feature information, enabling the model to denoise more thoroughly. Additionally, it utilizes Attention mechanism to calculate the weight coefficients between feature information, allowing the model to focus on key data features and reduce the distortion

of denoised signals. Finally, the feature information is reconstructed to obtain the denoised signal. We verify the denoising effectiveness of ADNet on both simulated and experimental datasets. The experimental results demonstrate that our proposed method outperforms traditional methods and recently proposed data-driven denoising methods. However, because the training of ADNet requires paired clean-noisy signals, which can be difficult to collect in industrial settings, we will investigate unsupervised neural network denoising methods based on the algorithm in this paper to reduce the need for clean-noisy data pairs and facilitate simpler application in industrial settings.

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